Faculty of Information Technology & Computer Science

Information Technology Department

INT433-Broadband Networks & Communication

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Question Set [1] Issued Feb 10, 2026

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Reading Assignment: The key here is to make sure that you are familiar with the basics of computer networks.

Question 1: Define broadband. What is significant about broadband systems?.

- It's a facility that is used to increase the speed of the communications
- It's a communication that is ON all the time
- It's a communication medium that carries multiple data/channels
- You need broadband provider or ISP to help setup the communication
- It offers a quick web surfing experience

Question 2: Which OSI layer is responsible for the following?

1. Determining the best path to route packets.
    a Network Layer
2. Providing end-to-end communications with reliable service.
    a Transport layer
3. Providing node-to-node communications with reliable service.
    a Data Link Layer

Question 3: Explain what is meant by 'telecommunication services' and 'applications' for network.

- telecommunication services
    - enables the exchange of information between users at different locations
- applications
    - Telephone
    - Cell
    - Web browsing

    - Instant Messaging
    - Short Message Service

Question 4: Explain the differences between a PSTN and ISDN.

- ISDN
    - Public networks are used for different services
        - Public switched telephone network
        - Private lines
        - Packet switched data networks
        - Circuit switched data networks
    - Users have different devices to connect, to the network
        - Computers/pcs
        - Terminals
        - Telephones
        - Mainframe computers
        - Private branch exchanges
- PSTN

Question 5: List the 5 layers in the internet protocol stack. What are the principal responsibilities of each of these layers?

- Application Layer
    - Responsible for the protocol number
    - Creating the message it self that is gonna be sent
- Transport Layer
    - Responsible for managing end-to-end communication reliably
        - TCP -> reliable
        - UDP -> not reliable -> streaming
- Network Layer
    - Responsible for managing the packet route entirely, from source to destination
        - from router to router – network to network – lan to lan
- Data Link Layer
    - responsible for managing the transferring process but from hop to hop, depending on the mac address and switchs
- Physical Layer
    - responsible for handling/managing the physical aspects of transformation like cables … etc

Question 6: Under time-division multiplexing how two or more conversion share the use of a common transmission facility?.

- Basically the time is divided into frames
- Each frame has equal size slots
- each circuit reserve a slot of the frame then establish the connection

Question 7: Prove that the number of bidirectional links needed in a complete connection of *n* hosts is

$$\frac{n(n-1)}{2}$$

- basically, each device, connected to each and every other device so its (n-1) connections
- so if we have n devices where each has (n-1) connections, then all the connections is n(n-1)
- but in this assumption, the mathematical representation counts the connection between the device and the other twice since connection from a to b,c,d is counted 3 times a-> b and a->c and a->d and the same connection happens when b -> a, c -> a and d -> a, so final connection count is (n(n-1))/2

Question 8: What are the advantages of packet switching?

- The message is split into packets for transmission
- Each packet has header, content, then trailer
- The transmission process happens following routing/forwarding mechanism in the network along some path node-to-node
- At each node, the packet is stored briefly then forwarded to the next hope -> store and forward mechanism
- Transmission is never interrupted
- No capacity is allocated for the packets

Question 9: True or false (circle the correct answer, T or F):

T F Time Division Multiplexing (TDM) is typically used in *analog* communications.

T F Statistical TDM is used when there is a good chance that *not* all users will be transmitting at once.

T F Packet-Switched networks can provide the same quality of service to real-time applications 9such as telephony) as *circuit switching* can.

T F A switch in a datagram network uses a routing table that is based on the *destination* address.

T F In datagram switching, there are *no* setup or teardown phases.